

# Data Structure and Programming, Spring 2025

## Written Homework #4

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Late policy:  $Score = Score \times 0.7^{(h/24)}$ ,  $h = \text{late hours} \leq 48$

### 1. (10%) AA Tree

What is the result of consecutively inserting 1, 2, 8, 5, 7, 3, 9, 4, 6, 10 into an empty AA tree? How many skew and split operations are performed? Please show the result of every intermediate step.

### 2. (15%) Red-Black Tree

- Draw the 2-4 tree representation of Figure 1.
- Show the result after inserting key 47 into the red-black tree in Figure 1. Use the top-down rules.
- Show the result after deleting key 16 from the red-black tree in Figure 1. Use the top-down rules.

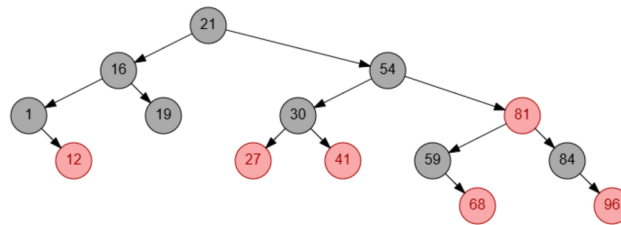


Figure 1: Red-Black Tree

3. (20%) **The Height of Red-Black Tree**

The height of a leaf is defined as 1.

- (a) Let  $M_h$  denote the maximum number of nodes in a red-black tree of height  $h$ . Find the recursive formula of  $M_h$ , and justify your answers.
- (b) Draw the minimum tree of a red-black tree with height 2, 3, 4 and 5.
- (c) Let  $N_h$  denote the minimum number of nodes in a red-black tree of height  $h$ . Find the recursive formula of  $N_h$ , and justify your answers.
- (d) According to 3(a) and 3(c), assuming that there are  $n$  nodes in the red-black tree, find the upper and lower bounds of the tree height.
- (e) Between AVL trees and Red-Black trees, which one enforces a stricter constraint on height? justify your answers.

4. (15%) **2-4 Tree**

- (a) After successively inserting 3, 2, 1, 5, 6, 8, 9, 7, 10, 4 into an empty 2-3-4 tree, what is the resulting tree? How many times does overflow occur? Please show the result of every intermediate step.
- (b) Following 4(a), what is the result of top-down deleting node 1 in the result 2-3-4 tree?

5. (9%) **Possible Red-Black Tree**

Suppose that we have numbers between 1 and 101 in a red-black tree with size  $n$  ( $1 \leq n \leq 101$ ) and we want to search for the number 76. Determine whether the following sequences could be or could not be the sequence of nodes examined and explain why or why not. (Hint: Red-black Tree is a balanced binary search tree)

- (a) 25 46 81 57 87 76
- (b) 89 18 44 80 50 77 76
- (c) 20 43 57 93 61 90 86 66 83 79 70 74 76

6. (16%) **Hash**

Consider successively inserting keys 93,54,13,7,87,44,60,10,9 into a hash table with 11 slots, where the hash function maps key  $x$  to  $h(x) = x \bmod 11$ . Show the hash tables using the following techniques:

- (a) Separate chaining.
- (b) Linear probing.
- (c) Double hashing with the second hash function defined as  $h_2(x) = 9 - (x \bmod 7)$ . In other words, the location of the  $i$ 'th probe is  $f_i(x) = h(x) + i \cdot h_2(x) \bmod 11$ .
- (d) Following Problem 6(b), remove key 87 and then insert key 88.

**7. (15%) Fill-in-the-blank questions**

Solve the following questions and justify your answers. You need to write down the process.

1. For a 2-3-4 tree of size 2025, let  $x$  be the minimum possible height,  $y$  be the maximum possible height, then  $(y - x) \bmod 5$  is \_\_\_\_.
2. For a red-black tree of height 5, let  $x$  be the minimum possible number of red nodes,  $y$  be the maximum possible number of red nodes, then  $(y - 2x) \bmod 5$  is \_\_\_\_.
3. For a binary heap of height 2025, let  $x$  be the minimum possible size,  $y$  be the maximum possible size, then  $(x + y) \bmod 5$  is \_\_\_\_.
4. Consider a binary min-heap that results from successively inserting keys 8,3,6,1,9,4,5,7,2 into an initially empty heap. What is the depth of key 5?
5. For a binary min-heap containing 2025 items, where the heap is represented as an array, and the smallest element is at index 0 (the root), show how to determine the possible positions of the 5th smallest element in the heap.